Call Your Shot!

Speaker notes, 60 minutes. Slides 1 to 19.

ARNAV operates the screen and the tool. **ASHISH** facilitates and keeps time. Gray italic lines are actions, not spoken.

Numbers to remember (measured in rehearsal, re-check on the day)

First shot (original, Across, 512)	**predicted 6.8%, actual 29.1%**
Collisions in the data	**about 3% of examples, error about 11x bigger**
Baseline (24 dev scenes)	**2/24, mean miss 0.385**
Collision weight 5x	**0/24, mean miss 0.545 (worse)**
Free-motion helper	**13/24, mean miss 0.101, no hit lost**
Baseline weights, Across	**predicts ~3%, actually misses ~55%**
Free-head weights, Across	**predicts 13.6%, actual 13.6%**
Diagonal, free-head	**still a miss, about 22%**
Free-head at 2,048 candidates	**Across lands, about 1.5%**
Real test: baseline	**6/96, mean miss 0.327**
Real test: free-motion	**69/96, mean miss 0.073; 63 fixed, 0 lost, 27 missed**

Our model, known geometry	**96/96 (mean miss 0.011)**
New approach, geometry inferred	**96/96 larger data; 94/96 original data only**
Found from data	**walls .0200/.9800, bumper (.5,.5) radius .1150, drag .985**
No-geometry plain network	**2/96 and 6/96**

If something goes wrong

Chat stalls over 60 s: "While that runs, who changed their mind?" Then run it in the terminal.

Running late, cut in order: stress case, worked example 1, one team prompt not two, the reveal. Never the real test.

Real test fails: say what completed, show the rehearsal numbers labelled as rehearsal.

Timeline

0:00 to 0:15	**Slides 1 to 8: check-in, intro, teams**
0:15 to 0:21	**Slides 9 to 10: the puzzle**
0:21 to 0:28	**Slides 11 to 13: diagnose**
0:28 to 0:36	**Slide 14: two worked examples**
0:36 to 0:41	**Slide 14: your idea (5 min)**
0:41 to 0:57	**Slide 15: team runs, real test, reveal**
0:57 to 1:00	**Slides 16 to 19: wrap up**

1 Check-in 0:00

ASHISH Hi everyone, welcome! Scan the code on screen and check in. It takes ten seconds. Then grab a seat with the people around you.

> *Arnav: open the site and ChatGPT Work. Check both load.*

2 Call Your Shot! 0:03

ASHISH Today's session is called Call Your Shot. Think, Build, Benchmark. By the end of the hour you'll have predicted what an AI model does, written your own prompt to fix it, and checked whether the fix really worked.

3 OpenAI Student Collective 0:05

ASHISH The Collective brings students together to try the newest AI tools, through workshops, challenges and campus events.

4 Meet your Campus Leads 0:06

ARNAV I'm Arnav, third year, chemistry and financial engineering. My favourite fun use of ChatGPT is decoding random internet lore.

ASHISH I'm Ashish, third year, chemistry. Mine is reading my notes for me.

5 Form a group 0:07

ASHISH Get into groups of six to seven. No pun intended. Introduce yourselves: name, major, year, hometown. Then tell the group something you've been unusually into lately.

ASHISH You're a team for the whole hour. Pick a discussion lead, a recorder and a spokesperson. The spokesperson casts your vote and hands in your prompt later.

> *Walk around for 90 seconds. "Hands up when you have a team of six or seven."*

6 ChatGPT Work

0:10

ARNAV ChatGPT Work is built for bigger jobs. Give it a plan and it can take on a whole project from start to finish. It connects to your tools and browser, and it can run on a schedule. Today we give it a bounded task, then check what it did.

7 Activity

0:12

ASHISH You'll direct ChatGPT Work to investigate a puck-planning model. Your team predicts, votes, and writes a prompt. We run it live. Scan this code to open the site. No login needed.

ARNAV We run the tool on the big screen, so nobody is blocked. You make the calls.

8 Puck Around, Find Out! 0:14

ARNAV This activity is called Puck Around, Find Out. Try things with AI. Keep what holds up.

9 The puzzle 0:15

ARNAV I used ChatGPT Work to plan a puck's path to a target in 35 steps, using a small neural network. Here's the problem.

> *Lab: Original neural network, Across, 512 candidates. Find a new shot.*

ASHISH Will the puck land in the green ring? Write yes or no. (30 seconds) Hands up for yes... and no.

> *Execute.*

ARNAV Predicted miss about 6.8 percent. Actual miss about 29.1 percent.

ASHISH Same push. The model predicted it would land right at the edge of the ring, and reality disagreed.

ASHISH A neural network is a small guesser. It sees where the puck is and how fast it's moving, and guesses where it will be one step later. Do that 35 times and you get the dashed path. No idea? Ask GPT for a few lines, or ask your neighbours.

ASHISH If you have a laptop, the Team kit tab has the files for later. Today we run everything live, so you don't need to install anything.

10 What you all should try 0:20

ASHISH Three things you can do with a model like this. Diagnose: find where prediction and reality split apart. Repair: change how the network is trained and compare. Stress-test: move the target, change the speed, add more search.

11 Choose your first idea 0:21

> *Paste the starter prompt into chat A NOW so the answer is ready by slide 12. It took about a minute in testing.*

ASHISH One minute in your team. What will the model get wrong first: collisions, long free motion, or the planner's picks? One hand per team.

> *Count. Take two 30-second reasons.*

12 Starter prompt 0:23

> *Show the answer.*

ARNAV Outcome, context, constraints, and a check. The constraint was: don't edit any files. It found the function that picks the push and the one that predicts paths.

> *Show the contact-vs-free table.*

ARNAV Collisions are about 3 percent of the training data, but the error there is about eleven times bigger.

ASHISH Which of your votes does that support? It supports it. It doesn't prove it.

> *If the chat stalls past 60 seconds: "While that runs, who changed their mind?" Then run it in the terminal.*

13 Discuss with your group 0:26

ASHISH Ninety seconds. What did the model predict, and what actually happened? Where do the paths differ? What one thing would you try, and what result would change your mind?

> *Walk the benches.*

> *Model answer if nobody volunteers: the paths split BEFORE any collision (5.5% gap by step 10). At step 11 the real puck clips the bumper; the predicted path passes it. The real puck then bounces off the bottom wall at step 30.*

14 Take your idea further 0:28

Worked example 1 (0:28)

> *Arnav runs the baseline.*

ASHISH All fixed, some better, or no change? Tick your sheet.

ARNAV The starter scores 2 out of 24, mean miss 0.385.

> *Run collision weight 5x.*

ARNAV Zero out of 24, mean miss 0.545. Worse.

ASHISH A sensible idea that made it worse. Keep that result. It tells us something, and we don't yet know why.

Worked example 2 (0:32)

ASHISH A different idea. Most of the path is gliding, which is simple and steady. So we give the gliding its own small learned helper.

> *Run the free-motion head.*

ARNAV 13 out of 24, mean miss 0.101, and no hit was lost. One scene's miss grew slightly, so we say that too.

> *Load the baseline weights, execute, then the free-head weights.*

ARNAV The starter predicted about a 3 percent miss and actually missed by 55. Now it predicts 13.6 and misses by 13.6. It's honest about what it can do.

Improvement 3: your idea (0:36)

ASHISH Improvement 3 is yours. Use your own brains. Pick ONE change. Say what you expect, and what result would change your mind. Write it as a prompt for ChatGPT Work: a goal, one change, what you keep fixed, and a check. Five minutes, hard stop. Tell us two things you would change.

ASHISH Stuck? Where did the paths split apart? What does the model see? How does it learn? How big is it?

> *Both hosts walk the benches: What do you expect? What would change your mind?*

15 Highlights 0:41

Time's up (0:41)

ASHISH Alright, time's up! Pens down. Spokespeople, tell us your prompts.

> *One sentence each. Pick two different ideas. Others: "Great idea, not today. We'll note it." Add the guardrail suffix. Arnav pastes into chat A, Ashish into chat B.*

While they run (0:42)

ASHISH While these run, let's break something.

> *Stress case: move the target to Diagonal.*

ARNAV With the repaired model, Diagonal is still a miss, about 22 percent. That's a limitation and we say it out loud.

> *Slider to 2,048 candidates, back on Across.*

ARNAV More search lands the shot now, about 1.5 percent, because the predictions are honest. These are chosen stress cases, not a robustness benchmark.

Let's see how it performed (0:48)

ASHISH Let's see how your ideas performed.

> *For each result: the team, the idea, hits, mean miss, what got better, what got worse. Credit the team. A worse result is a good result if we can explain it. If a chat is still working at 0:50: "We'll check back at the end."*

The real test (0:52)

ASHISH Now the real test. You picked your ideas using the 24 development scenes, so those scores are a bit optimistic. We score once on 96 scenes nobody tuned on. Freeze. No more edits.

> *Full puck-lab folder: python host/show_scores.py scratch/baseline.json scratch/candidate.json*

ARNAV Baseline: 6 out of 96, mean miss 0.327. Free-motion candidate: 69 out of 96, mean miss 0.073. 63 scenes fixed, no hits lost, 27 still missed.

ASHISH It helped on these scenes. It's one model, one seed, one simulator. We won't say robust.

How we got our good score (0:55)

ASHISH Quick vote: accept, reject, or qualify this change.

> *Hands. Accept = ship it. Reject = throw it out. Qualify = keep it, say where it does not hold.*

> *Switch to host view.*

ARNAV Our own model scored 96 out of 96 on these same scenes. It treats the puck gliding and the puck hitting something as two different problems. For hits, it describes each collision relative to the surface it hit, so a hit at any angle shares one learned response.

ASHISH Fair question: did we cheat by telling it where the walls and the bumper are? So we tested it. We threw away our hand-written numbers and let the data tell us. From the training data alone, it finds the walls and the bumper to four decimal places. The same model, built on those found numbers, still scored 96 out of 96. Using only the original small dataset it scored 94 out of 96.

ARNAV What we still assume is the shape of the table: straight walls and one round bumper. The numbers are learned. We changed several things at once, so we can't say which one mattered most, and it's one model in one simulator. But the lesson holds: look at where the model goes wrong, give that part the right structure, and test it fairly.

16 Check out 0:57

ASHISH Scan the code to share what you made with the OpenAI team and get a custom graphic. Three takeaways: confident isn't the same as correct, a sensible fix can fail so keep the result, and compare results including what got worse. ChatGPT Work can implement the change. Your job is to establish whether it helped.

17-18 How to stay connected 0:58

ARNAV Join the campus platform, follow our Luma page for events, and subscribe on Substack. Scan the codes on the next slide. Studio Hour details will be shared soon.

19 Fun fact

0:59

ASHISH Fun fact: all the elements in these slides were generated by GPT. Hope it looked good. Thanks, everyone!